

# LaRT v2.00 User Manual

## Contents

|                                                           |           |
|-----------------------------------------------------------|-----------|
| <b>1. Introduction</b>                                    | <b>2</b>  |
| <b>2. Installation and Build</b>                          | <b>3</b>  |
| 2.1 Prerequisites . . . . .                               | 3         |
| 2.2 Compilation . . . . .                                 | 3         |
| 2.3 Running . . . . .                                     | 3         |
| <b>3. Input File Format</b>                               | <b>4</b>  |
| <b>4. Grid Configuration</b>                              | <b>4</b>  |
| 4.1 Cartesian Grid . . . . .                              | 4         |
| 4.2 Grid Geometry . . . . .                               | 4         |
| 4.3 Symmetry and Periodic Boundaries . . . . .            | 5         |
| 4.4 Distance Units . . . . .                              | 5         |
| 4.5 Grid Type Selection . . . . .                         | 5         |
| <b>5. Source Configuration</b>                            | <b>5</b>  |
| 5.1 Source Geometry . . . . .                             | 5         |
| 5.2 Spectral Type . . . . .                               | 6         |
| 5.3 Line Identification . . . . .                         | 7         |
| 5.3.1 Combined Hydrogen + Deuterium Ly $\alpha$ . . . . . | 8         |
| 5.4 Frequency and Wavelength Grid . . . . .               | 9         |
| <b>6. Medium Properties</b>                               | <b>9</b>  |
| 6.1 Optical Depth Normalization . . . . .                 | 9         |
| 6.2 Temperature . . . . .                                 | 9         |
| 6.3 Density . . . . .                                     | 10        |
| 6.4 Velocity Fields . . . . .                             | 10        |
| <b>7. AMR Grid Mode</b>                                   | <b>11</b> |
| 7.1 Enabling AMR Mode . . . . .                           | 11        |
| 7.2 Generic AMR Data Format . . . . .                     | 11        |
| 7.3 Tau Normalization Convention . . . . .                | 12        |
| 7.4 Source Geometries in AMR Mode . . . . .               | 12        |
| <b>8. Clumpy Medium Mode</b>                              | <b>12</b> |
| 8.1 Overview . . . . .                                    | 12        |
| 8.2 Enabling Clump Mode . . . . .                         | 12        |

|            |                                                                      |           |
|------------|----------------------------------------------------------------------|-----------|
| 8.3        | Clump Parameters . . . . .                                           | 13        |
| 8.4        | Spatially-varying clump properties . . . . .                         | 14        |
| 8.5        | Standalone clump generator ( <code>make_clumps.x</code> ) . . . . .  | 14        |
| 8.6        | Overlapping Clumps . . . . .                                         | 15        |
| 8.7        | Clump Velocity . . . . .                                             | 16        |
| <b>9.</b>  | <b>Observers and Peel-off</b>                                        | <b>16</b> |
| 9.1        | Observer Placement . . . . .                                         | 16        |
| 9.2        | Multiple Observers . . . . .                                         | 17        |
| 9.3        | Image Parameters . . . . .                                           | 17        |
| 9.4        | Peel-off Output Flags . . . . .                                      | 17        |
| 9.5        | Sight-line Optical-depth and Column-density Maps . . . . .           | 17        |
| 9.6        | Inside Observer (HEALPix All-Sky) . . . . .                          | 19        |
| <b>10.</b> | <b>Output Files</b>                                                  | <b>20</b> |
| 10.1       | Primary Output Arrays . . . . .                                      | 20        |
| 10.2       | Angular Spectrum $J_{\mu}$ . . . . .                                 | 20        |
| 10.3       | Peel-off cube units and conversion to $J_{\mu}$ units . . . . .      | 21        |
| 10.4       | Output Control Parameters . . . . .                                  | 23        |
| <b>11.</b> | <b>Dust and Polarization</b>                                         | <b>23</b> |
| <b>12.</b> | <b>Parallelism</b>                                                   | <b>23</b> |
| <b>13.</b> | <b>Complete Parameter Reference</b>                                  | <b>24</b> |
| <b>14.</b> | <b>Python Utilities</b>                                              | <b>28</b> |
| 14.1       | Reading FITS output: <code>read_lart.py</code> . . . . .             | 28        |
| <b>15.</b> | <b>Example Input Files</b>                                           | <b>29</b> |
| 15.1       | Uniform Sphere, $\text{Ly}\alpha$ , Static . . . . .                 | 29        |
| 15.2       | Sphere with Hubble Outflow and Peel-off . . . . .                    | 30        |
| 15.3       | Realistic Galaxy (FITS Density Field, Stokes Polarization) . . . . . | 30        |
| 15.4       | AMR Sphere (Generic Text Format) . . . . .                           | 31        |
| 15.5       | Clumpy Sphere with Random Velocities . . . . .                       | 31        |

## 1. Introduction

**LaRT** (**L**ayman-**a**lpha **R**adiative **T**ransfer) is a Monte Carlo radiative transfer code for computing the propagation of resonance-line photons through astrophysical neutral-gas media. The default transition is Lyman- $\alpha$  ( $\text{Ly}\alpha$ ,  $\text{H I } 1216 \text{ \AA}$ ),

but many other UV/optical/IR lines are supported (see §5.3). Dust scattering and absorption, full Stokes polarization tracking, Hubble-flow and other bulk velocity fields, and spatially varying density, temperature, and velocity from external data files are all supported.

Starting from v2.00, the Cartesian grid is supplemented by:

- **AMR mode** — octree adaptive mesh refinement, allowing spatially varying resolution (§7.).
- **Clumpy medium mode** — a population of compact, optically thick spherical clumps embedded in a transparent inter-clump medium (§8.).

This manual covers v2.00 and v2.10. The only user-visible change in v2.10 is the introduction of `par%grid_type` as a unified selector for the three grid modes; the old `par%use_amr_grid` and `par%use_clump_medium` flags remain valid for backward compatibility.

## 2. Installation and Build

### 2.1 Prerequisites

- Fortran 2003+ compiler: Intel `ifort/ifx` or GNU `gfortran`  $\geq 7$ .
- CFITSIO library.
- MPI library (MPICH or OpenMPI).

### 2.2 Compilation

```
cd LaRT_v2.00      # or LaRT_v2.10
make               # produces LaRT.x  (pure MPI)
make DEBUG=1       # full debug flags + runtime checks
make F90=mpif90     # override compiler
make clean          # remove .o and .mod files
make cleanall       # also remove executables
make ramses2fits    # auxiliary RAMSES converter
```

Every make invocation recompiles all objects (no incremental builds).

### 2.3 Running

```
mpirun -np 8 LaRT.x input.in
```

Output is written to `<base>.fits.gz`, where `<base>` is the input file base name (without `.in`) unless `par%out_file` is set explicitly.

### 3. Input File Format

Input files use Fortran *namelist* syntax. All parameters are prefixed with `par%` and collected in a single `&parameters` block.

```

1 &parameters
2   par%no_photons      = 1e6
3   par%temperature     = 1.0e4      ! gas temperature [K]
4   par%taumax          = 1.0e4
5   par%nx = 101, par%ny = 101, par%nz = 101
6   par%rmax            = 1.0
7   par%source_geometry = 'point'
8   par%spectral_type   = 'voigt'
9   par%nxfreq          = 121
10 /

```

- String values must be enclosed in single quotes.
- Multiple parameters may appear on one line, separated by commas.
- Lines beginning with `!` are comments.
- Unspecified parameters retain compiled defaults (listed in §13.).

## 4. Grid Configuration

### 4.1 Cartesian Grid

The simulation domain is a rectangular box centered at the origin. By default  $x \in [-x_{\max}, +x_{\max}]$  (and similarly for  $y$  and  $z$ ). The cell size is  $\Delta x = 2x_{\max}/n_x$ .

| Parameter                                 | Default | Description                                        |
|-------------------------------------------|---------|----------------------------------------------------|
| <code>par%nx, par%ny, par%nz</code>       | 101     | Number of cells per axis                           |
| <code>par%xmax, par%ymax, par%zmax</code> | 1.0     | Half-box size per axis (code units)                |
| <code>par%rmax</code>                     | -999    | Spherical boundary: density = 0 for $r > r_{\max}$ |
| <code>par%rmin</code>                     | -999    | Inner cavity: density = 0 for $r < r_{\min}$       |

If `par%input_field` or `par%dens_file` is given,  $(n_x, n_y, n_z)$  and  $(x_{\max}, y_{\max}, z_{\max})$  are read from the FITS header automatically.

### 4.2 Grid Geometry

`par%geometry` selects the spatial symmetry used for accumulating and saving the mean-intensity array  $J$ :

| Value                  | $J$ output geometry             |
|------------------------|---------------------------------|
| 'sphere' (default)     | 1-D radial profile $J(r)$       |
| 'cylinder'             | 2-D cylindrical $J(\rho, z)$    |
| 'rectangle'            | 3-D Cartesian cube $J(x, y, z)$ |
| 'plane_atmosphere'     | 1-D slab $J(z)$                 |
| 'spherical_atmosphere' | 1-D spherical shell $J(r)$      |

### 4.3 Symmetry and Periodic Boundaries

| Parameter        | Default | Effect                                                          |
|------------------|---------|-----------------------------------------------------------------|
| par%xyz_symmetry | .false. | Simulate only the +x,+y,+z octant (8× RAM saving)               |
| par%xy_symmetry  | .false. | Simulate only the +x,+y quadrant (4× RAM saving)                |
| par%xy_periodic  | .false. | Periodic in x,y: photons exiting one face re-enter the opposite |

Symmetry options are disabled automatically when peel-off is active.

### 4.4 Distance Units

| Parameter         | Default | Description                                             |
|-------------------|---------|---------------------------------------------------------|
| par%distance_unit | "       | Physical unit: 'kpc', 'pc', 'au', or "" (dimensionless) |
| par%distance2cm   | 1.0     | Alternatively: 1 code unit in cm                        |

In dimensionless mode (par%distance\_unit = ""), the gas density is determined entirely by par%taumax and the grid geometry; no physical unit is needed.

### 4.5 Grid Type Selection

| Parameter                  | Default | Description                                            |
|----------------------------|---------|--------------------------------------------------------|
| par%grid_type <sup>†</sup> | 'car'   | <b>v2.10+</b> : 'car' (Cartesian), 'amr', or 'clump'   |
| par%use_amr_grid           | .false. | <b>v2.00</b> : enable AMR (deprecated in v2.10)        |
| par%use_clump_medium       | .false. | <b>v2.00</b> : enable clump mode (deprecated in v2.10) |

## 5. Source Configuration

### 5.1 Source Geometry

par%source\_geometry controls the spatial distribution of photon emission:

| Value                  | Description                                                                                               |
|------------------------|-----------------------------------------------------------------------------------------------------------|
| 'point'                | Point source at<br>(par%xs_point, par%ys_point, par%zs_point); default<br>(0,0,0)                         |
| 'uniform'              | Uniform volume emissivity throughout the grid (or inside<br>par%source_rmax)                              |
| 'uniform_sphere'       | Uniform within a sphere of radius par%source_rmax                                                         |
| 'uniform_xy'           | Uniform in the $z = 0$ plane                                                                              |
| 'gaussian'             | Gaussian: $\exp(-r^2/2\sigma_r^2)\exp(-z^2/2\sigma_z^2)$ ; widths<br>par%source_rscale, par%source_zscale |
| 'exponential'          | Exponential disc: $\exp(-r/r_s)\exp(- z /z_s)$ ; scales<br>par%source_rscale, par%source_zscale           |
| 'exponential_sphere'   | Exponential clipped to sphere of radius par%source_rmax                                                   |
| 'exponential_cylinder' | Exponential clipped to a cylinder                                                                         |
| 'sersic' / 'ssh'       | Sérsic profile; index par%sersic_m, half-light radius par%Reff                                            |
| 'diffuse_emissivity'   | 3-D emissivity read from par%emiss_file                                                                   |
| 'star_file'            | Individual point sources from text file par%star_file (columns:<br>$x, y, z, L$ )                         |

Associated parameters:

| Parameter                                | Default | Description                                  |
|------------------------------------------|---------|----------------------------------------------|
| par%xs_point, par%ys_point, par%zs_point | 0.0     | Point source position (code units)           |
| par%source_rmax                          | -999    | Outer truncation radius for extended sources |
| par%source_rscale                        | -999    | Radial scale length                          |
| par%source_zscale                        | 0.0     | Vertical scale height                        |
| par%sersic_m                             | 4.0     | Sérsic index                                 |
| par%Reff                                 | 1.0     | Effective (half-light) radius                |
| par%comoving_source                      | .true.  | Source photons co-moving with the local gas  |

## 5.2 Spectral Type

par%spectral\_type sets the frequency distribution of emitted photons:

| Value            | Description                                                                    |
|------------------|--------------------------------------------------------------------------------|
| 'monochromatic'  | Delta function at line center ( $x = 0$ )                                      |
| 'voigt'          | Voigt profile at the local gas temperature                                     |
| 'voigt0'         | Voigt profile at the fixed temperature <code>par%temperature0</code>           |
| 'gaussian'       | Gaussian with width <code>par%gaussian_width_vel</code> [ $\text{km s}^{-1}$ ] |
| 'continuum'      | Flat spectrum over [ <code>par%xfreq_min</code> , <code>par%xfreq_max</code> ] |
| 'line_prof_file' | Arbitrary profile read from <code>par%line_prof_file</code>                    |

### 5.3 Line Identification

The atomic transition is selected with `par%line_id` (default:  $\text{Ly}\alpha$ ).

| <code>par%line_id</code>  | Transition                     | Wavelength            |
|---------------------------|--------------------------------|-----------------------|
| (default)                 | H I $\text{Ly}\alpha$          | 1215.67 Å             |
| 'ly_alpha_HD'             | H I + D I $\text{Ly}\alpha$    | 1215.67 Å (see below) |
| 'CIV_1548'                | C IV doublet                   | 1548.19 / 1550.77 Å   |
| 'NV_1239'                 | N V doublet                    | 1238.82 / 1242.80 Å   |
| 'OVI_1032'                | O VI doublet                   | 1031.91 / 1037.61 Å   |
| 'SiIV_1394'               | Si IV doublet                  | 1393.76 / 1402.77 Å   |
| 'NaI_D'                   | Na I D doublet                 | 5891.58 / 5897.56 Å   |
| 'CaII_HK'                 | Ca II H & K doublet            | 3934.78 / 3969.59 Å   |
| 'MgII_2796'               | Mg II doublet                  | 2796.35 / 2803.53 Å   |
| 'AlII_1671'               | Al II singlet                  | 1670.79 Å             |
| 'SiII_1190' / 'SiII_1193' | Si II doublet (+ fluorescence) | 1190.42 / 1193.29 Å   |
| 'SiII_1260'               | Si II (+ fluorescence)         | 1260.42 Å             |
| 'SiII_1304'               | Si II (+ fluorescence)         | 1304.37 Å             |
| 'SiII_1527'               | Si II (+ fluorescence)         | 1526.71 Å             |
| 'FeII_2250'               | Fe II (+ fluorescence)         | 2249.88 Å             |
| 'FeII_2261'               | Fe II (+ fluorescence)         | 2260.78 Å             |
| 'FeII_UV3' / 'FeII_2344'  | Fe II UV3 multiplet            | 2344.21 Å             |
| 'FeII_UV2' / 'FeII_2383'  | Fe II UV2 multiplet            | 2382.76 Å             |
| 'FeII_UV1' / 'FeII_2600'  | Fe II UV1 multiplet            | 2600.17 Å             |
| 'HeI_10833'               | He I triplet                   | 10832.06 Å            |

The numerical `line%line_type` that controls the dispatch of `par%calc_voigt` / `par%do_resonance` (set internally by `setup_resonance_line`) takes the following values: 1 = singlet (one upward transition); 2 = doublet (two upward transitions sharing a single thermal width, e.g. C IV, N V, O VI, Na I D, Ca II H&K, Mg II, Si IV); 4 = one resonance plus one or more fluorescent decays (most Si II, Fe II UV3); 5 = multiple upward + multiple downward transitions, the most general case (Si II 1190/1193, Fe II UV1, Fe II UV2); 6 = three upward transitions plus one downward

transition (He I 10833); 7 = combined H + D Ly $\alpha$  (ly\_alpha\_HD, two coexisting two-level resonance scatterers, no fine structure).

### 5.3.1 Combined Hydrogen + Deuterium Ly $\alpha$

The line id 'ly\_alpha\_HD' treats neutral hydrogen and deuterium Ly $\alpha$  as two coexisting two-level resonance scatterers in the same medium. The two transitions sit only 0.33 Å ( $\sim 82$  km/s) apart and have nearly identical atomic constants except for the atomic mass and rest wavelength. The deuterium-to-hydrogen number-abundance ratio is set by

| Parameter             | Default              | Description                   |
|-----------------------|----------------------|-------------------------------|
| par%D_to_H_ratio      | $1.5 \times 10^{-5}$ | $n_D/n_H$ (cosmic primordial) |
| par%include_deuterium | .false.              | Convenience flag              |

Setting `par%include_deuterium = .true.` together with `par%line_id = 'ly_alpha'` promotes the run automatically to 'ly\_alpha\_HD', so existing input files can opt in without changing the line id.

Physical assumptions in this version (v1):

- Both H and D are treated as simple two-level resonance scatterers (`line_type=7`). Fine-structure splitting is *not* included for either species; if `par%fine_structure = .true.` is set together with 'ly\_alpha\_HD', the code emits a warning and disables fine structure.
- The phase function is the simple dipole ( $E_1 = 1, E_2 = 0, E_3 = 1$ ), identical for both species.
- Oscillator strengths and damping rates are taken to be equal for H and D ( $f_D = f_H, \Gamma_D = \Gamma_H$  to better than 0.1%). Mass and rest wavelength differences are fully accounted for in the Voigt evaluation and species selection.
- Photon `xfreq` is carried in H-frame Doppler units throughout. D-frame conversions happen on-the-fly in the combined Voigt and scattering routines via constants precomputed at line setup (`line%delta_nu_HD_Hz`, `line%ratio_Dfreq_HD`, `line%ratio_voigta_HD`). No per-cell deuterium arrays are stored.
- During a deuterium scattering event, the perpendicular thermal velocity components and the recoil shift are evaluated using the H atomic constants, which produces a small ( $\sim 1/\sqrt{2}$ ) error in the re-emission shape from D-scattered photons. Because D scatters are rare at cosmic abundance and the dominant D feature (the absorption dip) depends on opacity rather than re-emission shape, this is acceptable for v1. Per-photon species tracking is also not yet implemented.

The Deuterium Ly $\alpha$  rest wavelength used is  $\lambda_D = 1215.337$  (NIST vacuum),  $\sim 0.33$  blueward of  $\lambda_H = 1215.668$ . In LaRT's velocity convention ( $v = -c \Delta\lambda/\lambda_H$ , so negative  $v$  is blueshifted), the D Ly $\alpha$  line center sits at  $v_D \approx -82$  km/s. A reference example with five validation runs and a Jupyter notebook is provided under `examples/lya_HD/` (see `run.sh` and `inspect_HD.ipynb`). One of those runs is configured for direct comparison against Dijkstra, Haiman & Spaans (2006), ApJ 649, 14, Figure 3 ( $D/H = 3 \times 10^{-5}$ ,  $\tau \sim 10^6$ ).



## 5.4 Frequency and Wavelength Grid

Internally, photon frequencies are tracked in the dimensionless Doppler variable:

$$x = \frac{\nu - \nu_0}{\Delta\nu_D}, \quad \Delta\nu_D = \frac{\nu_0}{c} \sqrt{\frac{2kT}{m}}, \quad (1)$$

where  $\nu_0$  is the line-center frequency,  $T$  is the local gas temperature, and  $m$  is the atomic/ionic mass. Output spectra are binned onto a uniform grid in  $x$  by default. A wavelength or velocity grid can be requested instead:

| Parameter                              | Default | Description                                     |
|----------------------------------------|---------|-------------------------------------------------|
| par%nxfreq                             | 121     | Number of spectral bins                         |
| par%xfreq_min, par%xfreq_max           | auto    | Limits of $x$ -grid                             |
| par%xfreq0                             | 0.0     | Frequency offset applied to all emitted photons |
| par%nwavelength                        | 0       | Number of wavelength bins (overrides $x$ -grid) |
| par%wavelength_min, par%wavelength_max | —       | [Å]                                             |
| par%nvelocity                          | 0       | Number of velocity bins (overrides $x$ -grid)   |
| par%velocity_min, par%velocity_max     | —       | [km s <sup>-1</sup> ]                           |

The range is set automatically from the mean medium temperature when no limits are provided.

## 6. Medium Properties

### 6.1 Optical Depth Normalization

The overall opacity is scaled by one of the following (specify at most one):

| Parameter     | Description                                                                           |
|---------------|---------------------------------------------------------------------------------------|
| par%taumax    | Line-center $\tau_0$ from the grid center to the + $z$ boundary along the $z$ -axis   |
| par%tauhomo   | Same, assuming a uniform density field (scales an inhomogeneous field proportionally) |
| par%N_HI_max  | H I column density from center to + $z$ boundary [cm <sup>-2</sup> ]                  |
| par%N_HI_homo | Same, assuming uniform density                                                        |

For AMR grids, par%taumax is the pole-to-face optical depth (box center  $\rightarrow$  + $z$  face along  $z$ ), consistent with the Cartesian convention for a sphere of radius  $R = L/2$ .

### 6.2 Temperature

| Parameter        | Default           | Description                                                                              |
|------------------|-------------------|------------------------------------------------------------------------------------------|
| par%temperature  | 10 <sup>4</sup> K | Uniform gas temperature                                                                  |
| par%temperature0 | = $T$             | Temperature for spectral_type='voigt0' source                                            |
| par%bturb        | 0.0               | Turbulent velocity dispersion added in quadrature to thermal width [km s <sup>-1</sup> ] |
| par%temp_file    | "                 | 3-D temperature FITS cube [K]                                                            |

### 6.3 Density

**Analytic profiles.** Without external files, the density is set analytically via the opacity scale parameters (`par%taumax` etc.) combined with an optional radial profile:

| Profile type            | Key parameter                   | Functional form                                                       |
|-------------------------|---------------------------------|-----------------------------------------------------------------------|
| Uniform (sphere or box) | <code>par%rmax</code>           | Constant $n_{\text{H}}$ inside $r < r_{\text{max}}$                   |
| Spherical shell         | <code>par%rmin, par%rmax</code> | Constant $n_{\text{H}}$ between $r_{\text{min}}$ and $r_{\text{max}}$ |
| Exponential             | <code>par%density_rscale</code> | $n \propto \exp(-r/r_s)$                                              |

**External FITS data.**

| Parameter                    | Description                                                             |
|------------------------------|-------------------------------------------------------------------------|
| <code>par%dens_file</code>   | 3-D H I number density cube [ $\text{cm}^{-3}$ ]                        |
| <code>par%temp_file</code>   | 3-D temperature cube [K]                                                |
| <code>par%velo_file</code>   | 4-D velocity cube ( $v_x, v_y, v_z$ ) [ $\text{km s}^{-1}$ ]            |
| <code>par%emiss_file</code>  | 3-D emissivity cube [ $\text{photons s}^{-1} \text{cm}^{-3}$ ]          |
| <code>par%input_field</code> | Base name: reads <code>&lt;base&gt;.dens.fits.gz, .temp., .velo.</code> |

If `par%taumax` (or `par%N_HI_max`) is specified alongside a density file, the density is rescaled so that the target optical depth is achieved.

### 6.4 Velocity Fields

`par%velocity_type` selects the bulk velocity model. The default (empty string) is a static medium.

| Value                               | Key parameters                                                                | Velocity law                                                                                    |
|-------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <code>'hubble'</code>               | <code>par%Vexp</code>                                                         | $\mathbf{v}(r) = V_{\text{exp}} (r/r_{\text{max}}) \hat{r}$ ;<br>$V_{\text{exp}} > 0$ : outflow |
| <code>'constant_radial'</code>      | <code>par%Vexp</code>                                                         | $\mathbf{v}(r) = V_{\text{exp}} \hat{r}$ (constant magnitude)                                   |
| <code>'parallel_velocity'</code>    | <code>par%Vx, par%Vy, par%Vz</code>                                           | Uniform $\mathbf{v} = (V_x, V_y, V_z)$ [ $\text{km s}^{-1}$ ]                                   |
| <code>'ssh'</code>                  | <code>par%Vpeak,</code><br><code>par%rpeak,</code><br><code>par%DeltaV</code> | Bilinear profile (Song, Seon & Hwang 2020)                                                      |
| <code>'rotating_solid_body'</code>  | <code>par%Vrot, par%rinner</code>                                             | Solid-body rotation                                                                             |
| <code>'rotating_galaxy_halo'</code> | <code>par%Vrot, par%q</code>                                                  | Galaxy halo rotation curve                                                                      |

The `'ssh'` profile rises linearly to  $V_{\text{peak}}$  at  $r_{\text{peak}}$ , then shifts by  $\Delta V$  toward  $r_{\text{max}}$ :

$$v(r) = \begin{cases} V_{\text{peak}} r/r_{\text{peak}} & r < r_{\text{peak}} \\ \left[ V_{\text{peak}} + \Delta V \frac{r-r_{\text{peak}}}{r_{\text{max}}-r_{\text{peak}}} \right] \hat{r} & r \geq r_{\text{peak}} \end{cases} \quad (2)$$

A velocity FITS file (par%velo\_file) may be combined with a systematic par%velocity\_type.

Other velocity-related parameters:

| Parameter     | Default | Description                                                            |
|---------------|---------|------------------------------------------------------------------------|
| par%Omega     | 28.0    | Shear rate [(km s <sup>-1</sup> ) kpc <sup>-1</sup> ] for TIGRESS data |
| par%recoil    | .false. | Include photon-recoil frequency shift in scattering                    |
| par%core_skip | .false. | Enable core-skipping acceleration algorithm                            |

## 7. AMR Grid Mode

### 7.1 Enabling AMR Mode

**v2.00:**

```

1 par%use_amr_grid = .true.
2 par%amr_type     = 'generic' ! always specify explicitly
3 par%amr_file     = 'grid.dat'
```

**v2.10 (preferred):**

```

1 par%grid_type = 'amr'
2 par%amr_type  = 'generic'
3 par%amr_file  = 'grid.dat'
```

| Parameter       | Default  | Description                                            |
|-----------------|----------|--------------------------------------------------------|
| par%amr_type    | 'ramses' | Format: 'generic' (text) or 'ramses' (binary snapshot) |
| par%amr_file    | "        | Path to AMR data file or RAMSES snapshot directory     |
| par%amr_snapnum | 1        | Snapshot number for RAMSES output                      |

**Note:** the compiled default for par%amr\_type is 'ramses'. Always specify par%amr\_type = 'generic' explicitly when using the text format.

### 7.2 Generic AMR Data Format

Plain text: a header followed by one line per leaf cell.

```

1 # comment lines start with #
2 BOXLEN 100.0 ! box side length in code units
3 NLEAF 8192 ! number of leaf cells
4 ! columns: x y z level nH[cm^-3] T[K] vx[km/s] vy[km/s] vz[km/s]
5 50.5 50.5 50.5 1 1.0e-3 1.0e4 0.0 0.0 0.0
6 ...
```

Leaf-cell coordinates are given with the box origin at the corner, [0, BOXLEN]. The reader internally re-centers to  $[-L/2, +L/2]$ ; *do not* pre-shift the data.

### 7.3 Tau Normalization Convention

For AMR, `par%taumax` is the line-center optical depth from the box center to the  $+z$  face along the  $z$ -axis. This matches the Cartesian sphere convention where `par%taumax` is the center-to-surface optical depth, provided  $r_{\max} = L/2$ .

### 7.4 Source Geometries in AMR Mode

In v2.00, AMR supports `'point'` and `'uniform'`. In v2.10, most Cartesian geometries (Gaussian, exponential, Sérsic, `star_file`, etc.) are also available; refer to §5.1 and the v2.10 release notes.

## 8. Clumpy Medium Mode

### 8.1 Overview

Clumpy medium mode places  $N$  spherical clumps of radius  $r_c$  inside a spherical outer domain of radius  $R$ . The inter-clump medium is vacuum. All opacity resides inside the clumps. By default, clump positions are drawn by Random Sequential Adsorption (RSA) so that clumps never overlap. Optionally, overlapping clumps can be generated or loaded from an external FITS file; LaRT detects overlaps automatically after placement and engages an event-based multi-component raytrace in that case (see §8.6).

### 8.2 Enabling Clump Mode

**v2.00:**

```
1 par%use_clump_medium = .true.
```

**v2.10 (preferred):**

```
1 par%grid_type = 'clump'
```

### 8.3 Clump Parameters

| Parameter                            | Default              | Description                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>par%rmax</code>                | 1.0                  | Outer sphere radius (code units)                                                                                                                                                                                                                                                                                                           |
| <code>par%rmin</code>                | -999                 | Inner cavity radius (the default sentinel is treated as 0).<br>Clumps are placed only in the shell $r_{\min} \leq r \leq r_{\max}$                                                                                                                                                                                                         |
| <code>par%clump_radius</code>        | —                    | Individual clump radius (same units)                                                                                                                                                                                                                                                                                                       |
| <code>par%clump_tau0</code>          | —                    | Line-center $\tau_0$ from clump center to its surface                                                                                                                                                                                                                                                                                      |
| <code>par%clump_f_cov</code>         | —                    | Covering factor $C_f = Nr_c^2/R^2$                                                                                                                                                                                                                                                                                                         |
| <code>par%clump_f_vol</code>         | —                    | Volume filling factor $f_V = N(r_c/R)^3$                                                                                                                                                                                                                                                                                                   |
| <code>par%clump_N_clumps</code>      | —                    | Total number of clumps $N$                                                                                                                                                                                                                                                                                                                 |
| <code>par%temperature</code>         | $10^4$ K             | Clump gas temperature                                                                                                                                                                                                                                                                                                                      |
| <code>par%clump_sigma_v</code>       | 0.0                  | Gaussian $\sigma$ of random bulk clump velocities [km s <sup>-1</sup> ]                                                                                                                                                                                                                                                                    |
| <code>par%save_clump_info</code>     | <code>.false.</code> | Write clump positions and velocities to a FITS binary table                                                                                                                                                                                                                                                                                |
| <code>par%clump_input_file</code>    | “                    | If non-empty, load the clump population from this FITS file (see §8.5) instead of generating it internally                                                                                                                                                                                                                                 |
| <code>par%clump_fully_inside</code>  | <code>.true.</code>  | If <code>.true.</code> (default), reject any candidate whose outermost edge would protrude past the outer sphere ( $d_{\text{center}} + r_c > R$ ) or into the inner cavity ( $d_{\text{center}} - r_c < r_{\min}$ ); if <code>.false.</code> , restore the legacy behavior in which only the center is required to lie in $[r_{\min}, R]$ |
| <code>par%clump_allow_overlap</code> | <code>.false.</code> | If <code>.true.</code> , skip RSA overlap rejection during internal generation: clumps are placed at uniformly random positions. After placement <code>check_has_overlap()</code> is called automatically; if any overlapping pair is found the overlap-aware raytrace is engaged. See §8.6                                                |

Specify exactly one of `par%clump_f_cov`, `par%clump_f_vol`, or `par%clump_N_clumps`.

For the per-clump opacity, specify *one* of `par%clump_tau0`, `par%clump_NHI`, or `par%clump_nH`. As a convenience, the system-level radial-sightline targets `par%taumax` and `par%N_HI_max` are also accepted: when none of `par%clump_tau0`/`par%clump_NHI`/`par%clump_nH` is set, LaRT back-solves the peak per-clump opacity so the realised radial sightline from the box centre to  $r = R$  matches the requested `par%taumax` (or `par%N_HI_max`). In the uniform-clump case the back-solve uses the closed-form  $\tau_{\max} = \frac{4}{3} f_{\text{cov}} \tau_{c,0}$  (and similarly for  $N_{\text{HI,max}}$ ); in profile mode it uses the radial quadrature derived from `par%clump_radius_profile`, `par%clump_density_profile`, and `par%clump_number_profile`. `par%N_gasmax` is treated as an alias for `par%N_HI_max` in clump mode (HI-only line). When a pre-built clump population is loaded via `par%clump_input_file`, supplying `par%taumax` or `par%N_HI_max` multiplies every `cl_rhokap(i)` by a single factor  $\alpha$  so the realised target is matched (small-angle expected-column formula); the per-clump table is otherwise preserved.

When `par%rmin` is set, `par%clump_f_vol` and `par%clump_f_cov` are interpreted relative to the shell volume  $V_{\text{shell}} = \frac{4}{3}\pi(R^3 - r_{\text{min}}^3)$  and the shell sightline factor  $R^2 + Rr_{\text{min}} + r_{\text{min}}^2$ . With `par%rmin` unset (default -999, treated as 0) these reduce exactly to the original  $(R/r_c)^3$  and  $(R/r_c)^2$  formulas, preserving prior runs. When `par%clump_fully_inside` is on, the realized filling factors fall slightly below the requested target whenever  $r_c$  is a non-trivial fraction of  $R - r_{\text{min}}$ , since the available placement volume shrinks from  $\frac{4}{3}\pi(R^3 - r_{\text{min}}^3)$  to  $\frac{4}{3}\pi[(R - r_c)^3 - (r_{\text{min}} + r_c)^3]$ . Specify a slightly larger target if exact matching is required. See the *Clumpy Medium Algorithm Description* document, “Fully-Inside Constraint” subsection, for the placement details.

## 8.4 Spatially-varying clump properties

Three optional radial-profile inputs let the per-clump radius, internal HI density, and spatial number density vary with distance from the box center. Built-in shapes are ‘constant’ (default), ‘powerlaw’, ‘gaussian’, ‘exponential’, and ‘file’ (5-column ASCII table). Setting any flag to a value other than ‘constant’ switches `par%init_clumps` to the profile-aware path.

| Parameter                              | Default    | Description                                              |
|----------------------------------------|------------|----------------------------------------------------------|
| <code>par%clump_radius_profile</code>  | ‘constant’ | shape of $r_c(r)$                                        |
| <code>par%clump_radius_alpha</code>    | 0.0        | power-law index for $r_c(r)$                             |
| <code>par%clump_radius_r0</code>       | 0.0        | power-law scale for $r_c(r)$                             |
| <code>par%clump_density_profile</code> | ‘constant’ | shape of internal $n_H(r)$                               |
| <code>par%clump_density_alpha</code>   | 0.0        | power-law index for $n_H(r)$                             |
| <code>par%clump_density_r0</code>      | 0.0        | power-law scale for $n_H(r)$                             |
| <code>par%clump_number_profile</code>  | ‘constant’ | shape of spatial $n_{cl}(r)$                             |
| <code>par%clump_number_alpha</code>    | 0.0        | power-law index for $n_{cl}(r)$                          |
| <code>par%clump_number_r0</code>       | 0.0        | power-law scale for $n_{cl}(r)$                          |
| <code>par%clump_profile_file</code>    | “          | 5-column tabulated profile (used when any axis = ‘file’) |

For non-uniform profiles the closed-form  $f_{\text{cov}}/f_{\text{vol}}/N$  relation does not hold. Instead the profile-aware quadrature back-solves the spatial normalization so that whichever of `par%clump_f_cov`, `par%clump_f_vol`, or `par%clump_N_clumps` is provided is matched, and the other two are reported on stdout. See the *Clumpy Medium Algorithm Description* document for the integral definitions and the implementation outline.

## 8.5 Standalone clump generator (make\_clumps.x)

The same Fortran routines that build the clump population inside LaRT are also exposed through a small standalone driver, `make_clumps.x`:

```

1 make make_clumps                                # builds make_clumps.x
2 mpirun -np N ./make_clumps.x clumps.in          # writes <base>_clumps.fits.gz
```

The driver reads the same namelist as `LaRT.x` and writes the realized clump population to `<base_name>_clumps.fits.gz` without running the photon transport. Fix `par%iseed` to a non-zero value for reproducibility: under the same seed the table produced by `make_clumps.x` matches the `_clumps.fits.gz` written by a regular `LaRT.x` run with `par%save_clump_info = .true.` byte-for-byte.

To consume the file, set `par%clump_input_file` on the LaRT input:

```

1 &parameters
2   par%use_clump_medium = .true.
3   par%rmax              = 1.0
4   par%clump_radius      = 0.01
5   par%clump_tau0        = 1.0e3
6   par%temperature       = 1.0e4
7   par%no_photons        = 1e6
8   par%clump_input_file  = 'clumps_clumps.fits.gz'
9 /

```

When `par%clump_input_file` is non-empty, `init_clumps` skips the profile-detection and RSA-generation paths entirely and reads the population from the file. `read_clumps_fits` accepts files with any subset of the per-clump columns `R_CLUMP/RHOKAP/TEMP`; missing columns are filled from the corresponding header keyword (`CL_RAD/RHOKAP/TEMP_CL`). For the `RHOKAP` slot the column names `DENSITY` and `DENS` are also accepted as aliases, in priority order `RHOKAP` → `DENSITY` → `DENS`.

## 8.6 Overlapping Clumps

By default LaRT guarantees non-overlapping clumps through RSA placement and all raytrace paths assume this property. Two situations can produce overlapping populations:

1. A clump population loaded from an external FITS file via `par%clump_input_file` may contain overlapping clumps.
2. Setting `par%clump_allow_overlap = .true.` skips the RSA rejection step so that internally generated clumps are placed at uniformly random positions and will typically overlap whenever the clumps are large relative to the domain.

After the CSR acceleration grid is built, `check_has_overlap()` scans all clump pairs. If any overlapping pair is found it prints

```

1 Overlap check: overlapping clumps detected -- using overlap-aware raytrace.

```

sets an internal flag `has_overlap = .true.`, and routes all subsequent raytrace calls to the overlap-aware variants automatically. No input change is required; the detection and dispatch happen at run time.

### Physical model for overlapping clumps

Two clumps sharing a volume with different bulk velocities  $\mathbf{v}_1$  and  $\mathbf{v}_2$  are treated as two independent gas components coexisting at the same spatial location. The total line opacity at global-frame frequency  $x$  is

$$\kappa_{\text{total}}(x) = \sum_i \rho \kappa_i H(x - u_{\text{los},i}, a_i), \quad u_{\text{los},i} = \frac{\mathbf{v}_i \cdot \hat{\mathbf{k}}}{\Delta \nu_{D,i}}. \quad (3)$$

When a scattering event occurs in an overlap region the interacting atom belongs to one clump, chosen by opacity-weighted sampling  $P(\text{clump } i) = \kappa_i / \kappa_{\text{total}}$ .

### Quick-start example

Large clumps with low covering factor guarantee many overlaps while keeping the run fast:

```

1 par%use_clump_medium      = .true.      ! or par%grid_type = 'clump' (v2.10)
2 par%clump_radius          = 0.3
3 par%clump_f_cov           = 2.0
4 par%clump_allow_overlap   = .true.
5 par%no_photons            = 1e5
6 par%spectral_type         = 'voigt'
```

Reference inputs: `clump_overlap_mono.in`, `clump_overlap_sigv.in`, `clump_overlap_peel.in` in `examples/clump_sphere/`.

For a full algorithmic description see the *Clumpy Medium Algorithm Description* document, Section 15 “Overlap-Aware Raytrace”.

## 8.7 Clump Velocity

A systematic flow can be added on top of the random component (`par%clump_sigma_v`) via `par%velocity_type`. Supported values in clump mode: `'hubble'`, `'constant_radial'`, `'parallel_velocity'`, `'ssh'`, `'rotating_solid_body'`, `'rotating_galaxy_halo'`.

## 9. Observers and Peel-off

The peel-off (peeling-off) technique shoots a photon toward the observer at every scattering event, allowing spectral images to be constructed without noise. It is activated by setting `par%nxim` and `par%nyim` to non-zero values.

### 9.1 Observer Placement

**Direction vector:**

```

1 par%obsx = 0.0, par%obsy = 0.0, par%obsz = 1.0 ! look down +z axis
2 par%distance = 100.0                          ! observer-center separation
```

**Spherical angles:**



```

1 par%alpha = 45.0    ! azimuthal angle [deg]
2 par%beta  = 30.0    ! elevation angle [deg]
3 par%distance = 100.0

```

par%gamma sets the rotation of the detector plane about the line of sight. If par%distance is omitted it defaults to  $100 \times r_{\max}$  (or the maximum box half-extent).

## 9.2 Multiple Observers

Multiple observers are specified as arrays:

```

1 par%alpha = 0.0  45.0  90.0  135.0
2 par%beta  = 0.0   0.0   0.0   0.0

```

Up to 99 simultaneous observers are supported.

## 9.3 Image Parameters

| Parameter          | Default | Description                                                |
|--------------------|---------|------------------------------------------------------------|
| par%nxim, par%nyim | 0       | Image pixel counts; peel-off enabled when $> 0$            |
| par%dxim, par%dyim | auto    | Pixel angular size [deg] (defaults to cover the full grid) |

When par%dxim/par%dyim are not set, the pixel size is chosen to cover the full sphere:

$$\delta\theta = \frac{1}{N_x/2} \arcsin\left(\frac{r_{\max}}{d}\right). \quad (4)$$

## 9.4 Peel-off Output Flags

| Parameter              | Default | Description                                                                                                                                          |
|------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| par%save_peeloff       | .false. | Save 1-D peel-off spectra per observer                                                                                                               |
| par%save_peeloff_2D    | .false. | Save peel-off spectral image cubes ( $x$ - $y$ - $\nu$ )                                                                                             |
| par%save_peeloff_3D    | .false. | Save full 3-D peel-off data                                                                                                                          |
| par%save_sightline_tau | .false. | Compute deterministic sight-line maps from each observer pixel: gas $\tau(\nu)$ , $N_{\text{HI}}$ , and dust $\tau$ (when par%DGR $> 0$ ). See §9.5. |

## 9.5 Sight-line Optical-depth and Column-density Maps

When par%save\_sightline\_tau = .true., LaRT integrates the opacity along each observer pixel's line of sight and writes a separate FITS file <base>\_tau<obs>.fits.gz for every observer. The file contains:

| HDU      | Content                                                                                |
|----------|----------------------------------------------------------------------------------------|
| TAU_gas  | gas optical depth $\tau(\nu, x, y)$                                                    |
| N_gas    | gas column density $N_{\text{HI}}(x, y)$                                               |
| TAU_dust | dust optical depth $\tau_{\text{dust}}(x, y)$ (only when <code>par%DGR &gt; 0</code> ) |

The pixel grid uses the same WCS as the peel-off image (`par%nxim`, `par%nyim`, `par%dxim`, `par%dyim`); the frequency axis matches the spectrum output (`par%nxfreq`, `par%velocity_min`, `par%velocity_max`). All three grid modes (Cartesian, AMR, clump) are supported with the appropriate raytrace dispatch.

### Standalone driver `make_sightline_tau.x`

The same sight-line integration is exposed as a standalone executable that skips the photon-transport stage entirely:

```

1 make sightline_tau                                # builds make_sightline_tau.x
2 mpirun -np N ./make_sightline_tau.x sl.in          # writes <base>_tau.fits.gz

```

`sl.in` is an ordinary LaRT input file; the driver internally sets `par%save_peeloff` and the sight-line flag to `.true.` and otherwise honors every namelist parameter. Useful for parameter sweeps where only the deterministic maps are needed (no Monte-Carlo transport). Unlike `make_clumps.x`, multiple MPI ranks do speed this driver up: pixels are divided across `par%nproc` via `loop_divide` and reduced with `reduce_mem`.

The folder `examples/sightline_tau/` contains a reference set of six runs that exercise every grid mode and observer configuration the driver supports:

| File                            | Grid mode     | Observer                   |
|---------------------------------|---------------|----------------------------|
| <code>car_sightline.in</code>   | Cartesian     | outside (rectangular FITS) |
| <code>amr_sightline.in</code>   | AMR           | outside (rectangular FITS) |
| <code>clump_sightline.in</code> | clumpy sphere | outside (rectangular FITS) |
| <code>car_inside.in</code>      | Cartesian     | inside (HEALPix all-sky)   |
| <code>amr_inside.in</code>      | AMR           | inside (HEALPix all-sky)   |

The shipped `run.sh` executes all six in sequence; `make_amr_sphere_data.py` regenerates the AMR data file from a uniform-sphere physics model. The companion notebook `plot_sightline_tau.ipynb` displays the column-density maps, selected  $\tau(x, y, \nu)$  slices, the all-sky HEALPix mollview maps and the sky-mean line profile.

**tau\_huge cap (sight-line vs. photon transport).** The AMR raytrace routines used by the photon-transport stage early-exit when the running optical depth exceeds the double-precision  $\exp(-\tau)$  underflow point ( $\tau_{\text{huge}} \approx 745.2$ ); accumulating further contributes nothing to the photon's transmission. The sight-line variant `raytrace_to_edge_tau_gas_amr` exists precisely to *report* the integrated  $\tau$ , however large, and therefore runs without that cap (matching the Cartesian `raytrace_to_edge_tau_gas` in `raytrace_car.f90`). The same applies to `raytrace_to_dist_tau_gas_amr` used by the inside-observer HEALPix path.

## 9.6 Inside Observer (HEALPix All-Sky)

Setting `par%nside > 0` switches the entire output and peel-off chain to the *inside-observer* path: the observer is placed at a chosen position ( $o_x, o_y, o_z$ ) *inside* the medium, and each escaping photon is binned into an all-sky HEALPix map by the direction from the observer back to the photon. This mode is intended for “galactic” observations – e.g., placing the observer at the Sun’s position inside a model of the Galactic ISM, or inside a simulated galaxy.

### Activation:

```

1 par%nside = 8                ! HEALPix Nside parameter; total npix = 12*nside^2
2 par%obsx   = 0.3              ! observer position (code units)
3 par%obsy   = 0.0
4 par%obsz   = 0.0
5 par%save_peeloff = .true.
6 par%save_peeloff_2D = .true. ! also save the 2D map (sum over frequency)

```

When `par%nside > 0`, `setup.f90` automatically sets `par%observer_located_inside = .true.` and switches:

- the observer arrays from rectangular images to HEALPix all-sky maps with  $N_{\text{pix}} = 12N_{\text{side}}^2$ ;
- the peel-off routines to the `_inside` variants (which use `vec2pix` to bin the peeled weight by observer-pointing direction);
- `output_reduce`, `output_normalize` and `write_output` to the `_inside` variants.

In v2.10 the dispatch is performed by the type-bound methods of `grid_base_type` (`grid_class.f90`); the `grid_car_type` and `grid_amr_type` concrete types each branch on `par%observer_located_inside` and forward to the appropriate `_inside` or `_outside` concrete routine.

**Multiple observers.** As with the rectangular path, up to 99 observer positions are supported by passing arrays:

```

1 par%nside = 8
2 par%obsx   = 0.0  0.3  0.0
3 par%obsy   = 0.0  0.0  0.5
4 par%obsz   = 0.0  0.0  0.0

```

**Output files.** In inside-observer mode the `_obs.fits.gz` file holds 2D HEALPix arrays of shape `(npix, nxfreq)` with extensions `Direct` (unscattered direct contribution) and `Scattered` (scattered contribution). When `par%save_peeloff_2D = .true.` a separate `_obs2D.fits.gz` file is written with shape `(npix,)` holding the same arrays summed over frequency. The HEALPix maps use the RING ordering convention. The `Spectrum` HDU (`Jout`, `Jin`, `Jabs`) is always written regardless of observer mode.

**Source scope.** The inside-observer path supports the same internal sources as the standard path – `point`, `uniform`, `gaussian`, `exponential`, `sersic`, `diffuse_emissivity` – but *not* the external-illumination geometries (`point_illumination`, `stellar_illumination`, `plane_illumination`), which are only physically meaningful for an observer outside the medium. Stokes polarization is not yet implemented for the inside-observer path.

**Grid-mode support.** The HEALPix inside-observer path is available for both Cartesian (`par%grid_type = 'car'`) and AMR (`par%grid_type = 'amr'`) grids. In AMR mode, dedicated `_inside_amr` peeling, output and sightline routines are used (see the *LaRT AMR Description* document for implementation details). Clumpy medium (`par%grid_type = 'clump'`) is not currently supported with the inside-observer path.

**Sightline-tau output.** When the inside-observer path is active, an additional `_sightline.fits.gz` file is produced that records  $\tau$  and  $N(\text{HI})$  from the observer position to each of the six box faces ( $\pm x, \pm y, \pm z$ ). This is useful for cross-checking the optical depth structure seen by the observer.

**Smoke test.** The `examples/amr_sphere_generic/` directory ships `sphere_amr_inside_test.in` (AMR uniform sphere) and `sphere_car_inside_test.in` (Cartesian  $64^3$  counterpart), both with `par%nside = 8` and an off-center observer at  $(o_x, o_y, o_z) = (0.3, 0, 0)$ . These produce equivalent HEALPix maps within the documented AMR octree volume-averaging tolerance.

## 10. Output Files

All outputs are gzip-compressed FITS files (`.fits.gz`).

### 10.1 Primary Output Arrays

| Array | FITS extension | Description                                             |
|-------|----------------|---------------------------------------------------------|
| Jout  | JOUT           | Emergent spectrum integrated over all escape directions |
| Jin   | JIN            | Injected (intrinsic) source spectrum                    |
| Jabs  | JABS           | Spectrum of photons absorbed by dust                    |
| Jmu   | JMU            | Emergent spectrum binned by $\mu = \cos \theta_z$       |

**Jout** is the main science output. It is normalized per unit frequency interval and per unit area of the bounding sphere. In dimensionless units (no `par%distance_unit`):

$$J_{\text{out}}(x) = \frac{\text{escaped photon weight in bin } x}{n_{\text{ph}} \Delta x 2\pi R^2}, \quad (5)$$

where  $R = r_{\text{max}}$  (Cartesian) or  $R = L_{\text{box}}/2$  (AMR), and  $\Delta x = 1$  in units of the Doppler width.

**Jin** requires `par%save_Jin = .true.`

**Jabs** requires `par%save_Jabs = .true.`

**Jmu** requires `par%save_Jmu = .true.`; see §10.2.

The total flux is conserved:  $\int J_{\text{out}} dx + \int J_{\text{abs}} dx = \int J_{\text{in}} dx$  (in the absence of other losses).

### 10.2 Angular Spectrum Jmu

Setting `par%save_Jmu = .true.` adds a 2-D image extension `EXTNAME='Jmu'` of shape  $(n_x, n_\mu)$  that records the emergent spectrum binned by the  $z$ -component of the escape direction,  $\mu \equiv \cos \theta_z = \hat{k}_z$ . This is useful for

- verifying isotropy of homogeneous models (every  $\mu$  bin should match `Jout` within Poisson noise),
- characterizing the angular dependence of escape in non-isotropic models (for example, a slab or rotating halo).

**Binning.**

- `par%nmu` bins of width  $\Delta\mu$  tile the  $\mu$  range uniformly.
- Default `par%nmu = 11` (odd) places  $\mu = 0$  at a bin center, so the perpendicular-escape spectrum is preserved as a single channel. Even `par%nmu` places  $\mu = 0$  at a bin edge and cleanly separates the  $+z$  and  $-z$  hemispheres.
- Range:
  - `par%xyz_symmetry = .true.` folds escapes onto the  $+z$  hemisphere, so  $\mu \in [0, +1]$  and  $\Delta\mu = 1/n_\mu$ .
  - Otherwise  $\mu \in [-1, +1]$  and  $\Delta\mu = 2/n_\mu$ .

The header keywords `nmu`, `mu_min`, `dmu` record these values, and full WCS keywords (`CTYPE1='XFREQ'`, `CTYPE2='MU'`, `CRVAL*`, `CDELT*`) are written to the image header.

**Normalization.**  $J_{\mu}$  uses the same denominator as  $J_{\text{out}}$  multiplied by  $n_\mu$ , so that

$$\frac{1}{n_\mu} \sum_{i=1}^{n_\mu} J(x; \mu_i) = J_{\text{out}}(x) \quad (6)$$

holds exactly. Equivalently, in a homogeneous, isotropic medium each  $\mu$  bin reproduces  $J_{\text{out}}$  up to Poisson fluctuations, providing a cheap regression test for new geometries.

**Modes.** Available with all three grid modes: Cartesian, AMR octree, and the clumpy medium.

**10.3 Peel-off cube units and conversion to  $J_{\mu}$  units**

When peel-off is enabled (`par%nxim`, `par%nyim` > 0 and `par%save_peeloff_3D = .true.`), each observer's `<base>_obs_NNN.fits.gz` file contains `Scattered` and `Direct` 3-D image cubes, plus optionally `Direct0`. These are written by `output_sum_rect.f90` as a *specific intensity per pixel*, normalized so that

$$I_{\text{peel}}(x, y, \nu) = \frac{\text{photon weight in pixel } (x, y) \text{ and bin } \nu}{n_{\text{ph}} \delta\Omega_{\text{pix}} \Delta\nu d_{\text{cm}}^2}, \quad (7)$$

where  $\delta\Omega_{\text{pix}} = (\text{CD2\_2} \cdot \text{CD3\_3})(\pi/180)^2$  is the per-pixel solid angle,  $\Delta\nu$  is `Dxfreq` or `Dwave` (set by `par%intensity_unit`),  $d_{\text{cm}}^2$  is the square of the distance unit in cm, and  $n_{\text{ph}}$  is the photon count.

**Conversion to  $J_{\mu}$  units.**  $J_{\text{out}}$  and  $J_{\mu}$  are normalized by  $4\pi R^2$  (the source surface area) and the outward solid angle  $2\pi$ , whereas the peel-off cube is normalized by the per-pixel solid angle and  $d_{\text{cm}}^2$ . Summing the cube over pixels and rescaling brings the two onto a common axis:

$$J_{\text{peel}}(\nu) = \left( \sum_{(x,y)} I_{\text{peel}}(x, y, \nu) \right) \cdot \frac{4\pi D^2 \delta\Omega_{\text{pix}}}{A_{\text{surface}} \cdot 2\pi}, \quad (8)$$

where  $D$  is `DISTANCE` (in code units) and  $A_{\text{surface}}$  is the source surface area in code units, with the geometry-dependent forms mirroring `output_sum_rect.f90`:

| Geometry               | $A_{\text{surface}}$ (code units)                                                                  | Notes                                                                   |
|------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| sphere                 | $4\pi r_{\text{max}}^2$                                                                            | default; clumpy medium also uses this                                   |
| box (rectangle)        | $8(x_{\text{max}} y_{\text{max}} + y_{\text{max}} z_{\text{max}} + z_{\text{max}} x_{\text{max}})$ | full surface of $[-x_{\text{max}}, x_{\text{max}}] \times \dots$<br>box |
| slab (par%xy_periodic) | $2(2x_{\text{max}})(2y_{\text{max}})$                                                              | two unbounded boundaries                                                |
| AMR octree             | $4\pi$                                                                                             | equivalent to $r_{\text{max}} = 1$ (cf. §7.)                            |

The bin width  $\Delta\nu$  cancels because both  $J_{\text{mu}}$  and  $I_{\text{peel}}$  are already normalized per bin. At convergence and for an axisymmetric source,

$$J_{\text{peel}}(\nu; \alpha, \beta) \rightarrow J_{\text{mu}}(\nu; \mu = \cos \beta). \quad (9)$$

The Python helper `LaRTOutput.plot_peel_jmu_compare()` (in `python/read_lart.py`) implements Eq. (8) with the surface area chosen automatically from `par%geometry`, `par%rmax`, and the spectrum-header `XMAX/YMAX/ZMAX/XY_PER`. See also `examples/sphere/plot_Jpeel_conversion.ipynb` and `examples/clump_sphere/verify_peel_jmu.ipynb`.

**Component decomposition (clumpy medium).** For a point source inside a clumpy sphere with covering factor  $f_{\text{cov}}$ , the fraction of photons that escapes without scattering is  $\exp(-f_{\text{cov}})$  (Poisson statistics), and the fraction that scatters at least once is  $1 - \exp(-f_{\text{cov}})$ .  $J_{\text{mu}}$  tallies *both* populations. The peel-off cube splits them:

- **Scattered.** The MC estimator over many scatter events averages efficiently, so

$$\int J_{\text{peel}}^{\text{scatt}} d\nu \approx (1 - e^{-f_{\text{cov}}}) \int J_{\text{mu}} d\nu, \quad (10)$$

and the equality holds across all observer angles within Monte-Carlo noise.

- **Direct.** For a *point* source, every photon shares the same line of sight to a given observer, so the direct contribution is the binary realization of  $\exp(-\tau_{\text{LOS}})$ : either  $\sim 1$  (clear LOS) or  $\sim 0$  (blocked by an intervening clump) per observer.  $J_{\text{mu}}$ , by contrast, is azimuth averaged and captures direct escapes uniformly. Adding the binary `Direct` and the smooth `Scattered` therefore only matches  $J_{\text{mu}}$  *in expectation* – it can deviate substantially per observer at finite photon counts.

For shape comparisons in clumpy runs, prefer the scattered component (`component='scatt'` in `plot_peel_jmu_compare()`) so that Eq. (10) provides a clean, deterministic relation.

## 10.4 Output Control Parameters

| Parameter               | Default | Description                                            |
|-------------------------|---------|--------------------------------------------------------|
| par%out_file            | "       | Output file name (defaults to <input>.fits.gz)         |
| par%out_merge           | .false. | Accumulate into existing output instead of overwriting |
| par%out_bitpix          | -32     | FITS precision: -32 = float32, -64 = float64           |
| par%save_Jin            | .false. | Save injected spectrum                                 |
| par%save_Jabs           | .false. | Save dust-absorbed spectrum                            |
| par%save_Jmu            | .false. | Save angular spectrum $J(x; \mu)$ (§10.2)              |
| par%nmu                 | 11      | Number of $\mu$ bins for Jmu                           |
| par%save_all            | .false. | Save full 3-D $J(\nu, x, y, z)$ array                  |
| par%save_direc0         | .false. | Save photon escape direction distributions             |
| par%save_radial_profile | .false. | Save 1-D radial $J$ profile                            |
| par%save_sightline_tau  | .false. | Save optical depth along sightlines                    |
| par%intensity_unit      | "       | Physical unit label written to the FITS header         |

par%out\_merge = .true. is useful for noise reduction: run the same setup multiple times and merge the results. All input parameters must be identical across runs.

## 11. Dust and Polarization

| Parameter           | Default | Description                                                           |
|---------------------|---------|-----------------------------------------------------------------------|
| par%DGR             | 0.0     | Dust-to-gas ratio relative to the Milky Way value                     |
| par%albedo          | 0.32536 | Single-scattering albedo $\omega$                                     |
| par%hgg             | 0.67617 | Henyeey–Greenstein asymmetry parameter $g$                            |
| par%use_stokes      | .false. | Enable full Stokes polarization RT                                    |
| par%scatt_mat_file  | "       | Mueller matrix file for polarized dust scattering                     |
| par%use_reduced_wgt | .false. | Reduce photon weight by albedo instead of destroying absorbed photons |

When par%DGR = 0 (default), there is no dust and all photon weight is conserved. Non-zero par%DGR adds dust extinction and Henyeey–Greenstein scattering. When par%use\_stokes = .true., a Mueller matrix file must be provided; files for the Weingartner–Draine Milky Way model are included in the data/ directory.

## 12. Parallelism

| Parameter            | Default | Description                                                  |
|----------------------|---------|--------------------------------------------------------------|
| par%no_photons       | $10^6$  | Total photon packets (floating-point, converted to int64)    |
| par%use_master_slave | .true.  | Dynamic load balancing via master–worker algorithm           |
| par%num_send_at_once | 100     | Photons dispatched to each worker per batch                  |
| par%nprint           | $10^5$  | Progress-report interval [photons]                           |
| par%iseed            | auto    | Random seed; defaults derived from MPI rank and system clock |

With `par%use_master_slave = .true.`, one MPI rank acts as dispatcher and the rest as workers. This balances workloads when scattering rates vary across photon packets. With `.false.`, each rank processes an equal share of `par%no_photons`.

## 13. Complete Parameter Reference

Table 1 lists commonly used parameters. Parameters marked † are new or changed in v2.10.

Table 1: LaRT parameter reference.

| Parameter                                                                             | Default  | Description                       |
|---------------------------------------------------------------------------------------|----------|-----------------------------------|
| <i>Simulation control</i>                                                             |          |                                   |
| <code>par%no_photons</code>                                                           | $10^6$   | Number of photon packets          |
| <code>par%nprint</code>                                                               | $10^5$   | Progress interval [photons]       |
| <code>par%iseed</code>                                                                | auto     | Random seed                       |
| <i>Cartesian grid</i>                                                                 |          |                                   |
| <code>par%nx, par%ny, par%nz</code>                                                   | 101      | Cells per axis                    |
| <code>par%xmax, par%ymax,</code><br><code>par%zmax</code>                             | 1.0      | Half-box size (code units)        |
| <code>par%rmax</code>                                                                 | -999     | Spherical boundary radius         |
| <code>par%rmin</code>                                                                 | -999     | Inner cavity radius               |
| <code>par%geometry</code>                                                             | 'sphere' | $J$ output geometry (§4.2)        |
| <code>par%xyz_symmetry</code>                                                         | .false.  | +x, +y, +z octant only            |
| <code>par%xy_symmetry</code>                                                          | .false.  | +x, +y quadrant only              |
| <code>par%xy_periodic</code>                                                          | .false.  | Periodic in $x, y$                |
| <code>par%distance_unit</code>                                                        | "        | Physical unit ('kpc', 'pc', 'au') |
| <code>par%distance2cm</code>                                                          | 1.0      | 1 code unit in cm                 |
| <i>Grid mode</i>                                                                      |          |                                   |
| <code>par%grid_type</code> †                                                          | 'car'    | v2.10+: 'car', 'amr', 'clump'     |
| <code>par%use_amr_grid</code>                                                         | .false.  | v2.00 AMR flag                    |
| <code>par%use_clump_medium</code>                                                     | .false.  | v2.00 clump flag                  |
| <i>Source</i>                                                                         |          |                                   |
| <code>par%source_geometry</code>                                                      | 'point'  | See §5.1                          |
| <code>par%xs_point,</code><br><code>par%ys_point,</code><br><code>par%zs_point</code> | 0.0      | Point source position             |

*continued on next page*



(continued)

| Parameter                        | Default           | Description                                                 |
|----------------------------------|-------------------|-------------------------------------------------------------|
| par%source_rmax                  | -999              | Source truncation radius                                    |
| par%source_rscale                | -999              | Radial scale length                                         |
| par%source_zscale                | 0.0               | Vertical scale height                                       |
| par%sersic_m                     | 4.0               | Sérsic index                                                |
| par%Reff                         | 1.0               | Effective radius                                            |
| par%comoving_source              | .true.            | Source co-moving with local gas                             |
| par%emiss_file                   | "                 | Emissivity FITS cube                                        |
| par%star_file                    | "                 | Point-star list ( $x, y, z, L$ )                            |
| <i>Line and spectrum</i>         |                   |                                                             |
| par%line_id                      | Ly $\alpha$       | Atomic transition (§5.3)                                    |
| par%spectral_type                | 'monochromatic'   | Emission frequency distribution                             |
| par%temperature                  | 10 <sup>4</sup> K | Gas temperature                                             |
| par%temperature0                 | = $T$             | Source temperature for voigt0                               |
| par%bturb                        | 0.0               | Turbulent dispersion [km s <sup>-1</sup> ]                  |
| par%gaussian_width_vel           | 0.0               | Gaussian line width [km s <sup>-1</sup> ]                   |
| par%xfreq0                       | 0.0               | Frequency offset for emitted photons                        |
| par%line_prof_file               | "                 | Custom line profile file                                    |
| par%line_prof_file_type          | 0                 | 0: ( $x, I$ ); 1: ( $\lambda, I$ )                          |
| <i>Frequency grid</i>            |                   |                                                             |
| par%nxfreq                       | 121               | Spectral bins                                               |
| par%xfreq_min,                   | auto              | $x$ -grid limits                                            |
| par%xfreq_max                    |                   |                                                             |
| par%nwavelength                  | 0                 | Wavelength bins (overrides $x$ -grid)                       |
| par%wavelength_min,              | —                 | [Å]                                                         |
| par%wavelength_max               |                   |                                                             |
| par%nvelocity                    | 0                 | Velocity bins (overrides $x$ -grid)                         |
| par%velocity_min,                | —                 | [km s <sup>-1</sup> ]                                       |
| par%velocity_max                 |                   |                                                             |
| <i>Optical depth and density</i> |                   |                                                             |
| par%taumax                       | —                 | Line-center $\tau_0$ : center $\rightarrow +z$ boundary     |
| par%tauhomo                      | —                 | Same, uniform-density equivalent                            |
| par%N_HI_max                     | —                 | Column density: center $\rightarrow +z$ [cm <sup>-2</sup> ] |
| par%N_HI_homo                    | —                 | Same, uniform-density equivalent                            |

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(continued)

| Parameter                     | Default  | Description                                       |
|-------------------------------|----------|---------------------------------------------------|
| par%density_rscale            | -999     | Exponential density scale radius                  |
| <i>External data</i>          |          |                                                   |
| par%dens_file                 | "        | Density FITS cube [ $\text{cm}^{-3}$ ]            |
| par%temp_file                 | "        | Temperature FITS cube [K]                         |
| par%velo_file                 | "        | Velocity FITS cube [ $\text{km s}^{-1}$ ]         |
| par%input_field               | "        | Base name for<br>.dens/.temp/.velo.fits.gz        |
| <i>Velocity</i>               |          |                                                   |
| par%velocity_type             | "        | Velocity law (§6.4)                               |
| par%Vexp                      | 0.0      | Expansion/infall speed [ $\text{km s}^{-1}$ ]     |
| par%Vx, par%Vy, par%Vz        | 0.0      | Uniform velocity [ $\text{km s}^{-1}$ ]           |
| par%Vpeak                     | 0.0      | Peak velocity for SSH [ $\text{km s}^{-1}$ ]      |
| par%rpeak                     | 0.0      | Radius at peak for SSH                            |
| par%DeltaV                    | 0.0      | Velocity increment for SSH [ $\text{km s}^{-1}$ ] |
| par%Vrot                      | 0.0      | Rotation speed [ $\text{km s}^{-1}$ ]             |
| par%recoil                    | .false.  | Photon recoil in scattering                       |
| <i>AMR</i>                    |          |                                                   |
| par%amr_type                  | 'ramses' | 'generic' or 'ramses'                             |
| par%amr_file                  | "        | AMR data path                                     |
| par%amr_snapnum               | 1        | RAMSES snapshot number                            |
| <i>Clumpy medium</i>          |          |                                                   |
| par%clump_radius              | —        | Clump radius                                      |
| par%clump_tau0                | —        | Line-center $\tau_0$ per clump                    |
| par%clump_f_cov               | —        | Covering factor $C_f$                             |
| par%clump_f_vol               | —        | Volume filling factor $f_v$                       |
| par%clump_N_clumps            | —        | Number of clumps $N$                              |
| par%clump_sigma_v             | 0.0      | Random velocity dispersion [ $\text{km s}^{-1}$ ] |
| par%save_clump_info           | .false.  | Write clump table to FITS                         |
| <i>Observers and peel-off</i> |          |                                                   |
| par%nxim, par%nyim            | 0        | Image pixels; peel-off active when<br>> 0         |
| par%dxim, par%dyim            | auto     | Pixel size [deg]                                  |

continued on next page

*(continued)*

| Parameter                       | Default        | Description                                                          |
|---------------------------------|----------------|----------------------------------------------------------------------|
| par%distance                    | $100 r_{\max}$ | Observer–center distance                                             |
| par%obsx, par%obsy,<br>par%obsz | (0, 0, 1)      | Observer direction                                                   |
| par%alpha, par%beta             | —              | Azimuth, elevation [deg]                                             |
| par%gamma                       | 0.0            | Detector-plane rotation [deg]                                        |
| par%save_peeloff                | .false.        | 1-D peel-off spectra                                                 |
| par%save_peeloff_2D             | .false.        | Peel-off spectral images                                             |
| par%save_peeloff_3D             | .false.        | Full 3-D peel-off cube                                               |
| par%save_direc0                 | .false.        | Peel-off image without $e^{-\tau}$<br>(transparent-medium reference) |
| <i>Output</i>                   |                |                                                                      |
| par%out_file                    | "              | Output FITS file name                                                |
| par%out_merge                   | .false.        | Accumulate into existing file                                        |
| par%out_bitpix                  | −32            | FITS precision                                                       |
| par%save_Jin                    | .false.        | Save injected spectrum                                               |
| par%save_Jabs                   | .false.        | Save absorbed spectrum                                               |
| par%save_Jmu                    | .false.        | Save angular spectrum $J(x; \mu)$                                    |
| par%nmu                         | 11             | Number of $\mu$ bins for Jmu                                         |
| par%save_all                    | .false.        | Save full 3-D $J(\nu, x, y, z)$                                      |
| par%save_radial_profile         | .false.        | Save 1-D radial $J$                                                  |
| par%save_sightline_tau          | .false.        | Save sightline $\tau$                                                |
| par%intensity_unit              | "              | FITS header unit label                                               |
| <i>Dust</i>                     |                |                                                                      |
| par%DGR                         | 0.0            | Dust-to-gas ratio                                                    |
| par%albedo                      | 0.32536        | Single-scattering albedo                                             |
| par%hgg                         | 0.67617        | Henyey–Greenstein asymmetry $g$                                      |
| par%use_stokes                  | .false.        | Polarization RT                                                      |
| par%scatt_mat_file              | "              | Mueller matrix file                                                  |
| par%use_reduced_wgt             | .false.        | Reduce weight instead of destroy                                     |
| <i>Parallelism</i>              |                |                                                                      |
| par%use_master_slave            | .true.         | Dynamic load balancing                                               |
| par%num_send_at_once            | 100            | Photons per worker batch                                             |

## 14. Python Utilities

A small set of post-processing scripts is shipped under `python/` in each version's source tree. They depend on `numpy`, `astropy`, and (for plotting scripts) `matplotlib`.

### 14.1 Reading FITS output: `read_lart.py`

`read_lart.py` is a thin reader that loads the spectral arrays from a LaRT FITS output. It accepts the input-file name in any of three forms:

- full input file: `t1tau2.in`
- stem only: `t1tau2` (`.in` is added automatically)
- the FITS file: `t1tau2.fits.gz`

The script honors `par%out_file` when set in the namelist; otherwise it falls back to `<stem>.fits.gz` in the same directory as the input file.

#### Module API.

```

1 from read_lart import read_lart
2 out = read_lart('t1tau2.in')
3
4 # 1-D arrays, length nxfreq
5 out.xfreq, out.velocity, out.wavelength
6 out.Jout          # always present
7 out.Jin, out.Jabs, out.Jabs2    # optional -- None if not in file
8
9 # Jmu output -- present only if par%save_Jmu = .true. was used
10 out.mu           # bin centers, length nmu
11 out.mu_edges     # bin edges, length nmu+1
12 out.Jmu          # shape (nmu, nxfreq); None if absent
13 out.nmu, out.mu_min, out.dmu
14
15 # Raw FITS headers
16 out.spectrum_header, out.jmu_header

```

Every optional array is `None` when its FITS column or HDU is not present, so the reader works on outputs from any LaRT version.

**Plotting Jmu.** `LaRTOutput.plot_jmu()` draws the angular spectrum directly:

```

1 out = read_lart('t1tau3.in')
2 out.plot_jmu(kind='lines', x='velocity') # one curve per mu bin
3 out.plot_jmu(kind='image', x='velocity') # 2-D heatmap (x vs mu)

```

The `x=` keyword selects the abscissa ('velocity', 'xfreq', or 'wavelength'); the `kind=` keyword toggles between line ensemble (colored by  $\mu$ ) and a 2-D image. The line view also overplots  $J_{\text{out}}$  as a black dashed reference. Pass `ax=` to draw onto an existing axes, `savefig='out.png'` to save to disk. Axis limits accept either the tuple form `xlim=(lo,hi)` / `ylim=(lo,hi)` or the per-bound shorthand `xmin/xmax/ymin/ymax`; the per-bound override takes precedence when both are given:

```

1 out.plot_jmu(kind='lines', xlim=(-500, 500), ymin=0)
2 out.plot_jmu(kind='lines', xmax=300) # left edge auto-scaled

```

### Command-line usage.

```

1 python read_lart.py t1tau2 # auto-resolves to t1tau2.in
2 python read_lart.py t1tau2.in
3 python read_lart.py t1tau2.fits.gz

```

prints a one-screen summary of which arrays are present and, when `Jmu` is in the file, the per-bin  $\max_x J(x; \mu) / \max_x J_{\text{out}}(x)$  ratio (a quick isotropy sanity check for homogeneous models).

**Other scripts.** The following plotting helpers are also shipped:

- `plot_spectrum.py`, `plot_map.py`, `plot_intensity_profile.py`, `plot_Pa.py`: standard quick-look plots.
- `check_flux.py`, `flux_check.py`: flux-conservation diagnostics.
- `AMR_grid/`: helpers for generating generic AMR data files (see `AMR_grid/make_amr_grid.py`, `make_amr_sphere_radial.py`).

## 15. Example Input Files

### 15.1 Uniform Sphere, $\text{Ly}\alpha$ , Static

```

1 &parameters
2 par%no_photons      = 1e6
3 par%temperature     = 1.0e4
4 par%taumax          = 1.0e4
5 par%rmax             = 1.0

```

```

6 par%nx = 101, par%ny = 101, par%nz = 101
7 par%source_geometry = 'point'
8 par%spectral_type = 'voigt'
9 par%nxfreq = 121
10 /

```

## 15.2 Sphere with Hubble Outflow and Peel-off

```

1 &parameters
2 par%no_photons = 1e6
3 par%temperature = 1.0e4
4 par%taumax = 1.0e4
5 par%rmax = 1.0
6 par%nx = 101, par%ny = 101, par%nz = 101
7 par%source_geometry = 'point'
8 par%spectral_type = 'voigt'
9 par%velocity_type = 'hubble'
10 par%Vexp = 200.0 ! km/s outflow
11 par%nxfreq = 121
12 par%nxim = 129, par%nyim = 129
13 par%alpha = 0.0 45.0 90.0
14 par%beta = 0.0 0.0 0.0
15 par%save_peeloff_2D = .true.
16 /

```

## 15.3 Realistic Galaxy (FITS Density Field, Stokes Polarization)

```

1 &parameters
2 par%no_photons = 1e8
3 par%dens_file = 'galaxy.dens.fits.gz'
4 par%temp_file = 'galaxy.temp.fits.gz'
5 par%velo_file = 'galaxy.velo.fits.gz'
6 par%taumax = 1.0e4 ! rescale to this optical depth
7 par%distance_unit = 'kpc'
8 par%source_geometry = 'exponential'
9 par%source_rscale = 3.0
10 par%source_zscale = 0.5
11 par%spectral_type = 'voigt'
12 par%DGR = 1.0
13 par%use_stokes = .true.

```

```

14 par%nxfreq      = 121
15 par%nxim = 200, par%nyim = 200
16 par%save_peeloff_2D = .true.
17 /

```

## 15.4 AMR Sphere (Generic Text Format)

```

1 &parameters
2 par%use_amr_grid   = .true.           ! or par%grid_type = 'amr' in v2.10
3 par%amr_type       = 'generic'
4 par%amr_file       = 'sphere_amr.dat'
5 par%no_photons     = 1e6
6 par%taumax        = 1.0e4
7 par%source_geometry = 'point'
8 par%spectral_type  = 'voigt'
9 par%nxfreq        = 121
10 par%nxim = 100, par%nyim = 100
11 par%save_peeloff_2D = .true.
12 par%save_Jin      = .true.
13 /

```

## 15.5 Clumpy Sphere with Random Velocities

```

1 &parameters
2 par%use_clump_medium = .true.         ! or par%grid_type = 'clump' in v2.10
3 par%rmax             = 1.0
4 par%clump_radius     = 0.01
5 par%clump_f_cov      = 5.0
6 par%clump_tau0       = 1.0e3
7 par%temperature      = 1.0e4
8 par%clump_sigma_v    = 50.0           ! km/s random dispersion
9 par%no_photons       = 1e6
10 par%spectral_type    = 'voigt'
11 par%nxfreq          = 301
12 par%velocity_min     = -200.0
13 par%velocity_max     = 200.0
14 par%save_clump_info  = .true.
15 /

```